redblinkinglight at PAN 2026: Stylometric Feature Engineering and Gradient Boosting for Multi-Author Writing Style Analysis

Notebook for the PAN Lab at CLEF 2026

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Abstract

This paper describes an exploratory system for sentence-level author switch detection, developed for the PAN 2026 shared task at CLEF. The task requires identifying sentence boundaries where writing style changes within multi-author documents. Rather than using a transformer-based approach, this work builds a lightweight, interpretable system based on a 12,353-dimensional stylometric representation comprising lexical, punctuation, function-word, character n-gram, part-of-speech, and language-model features. Consecutive sentence pairs are represented as absolute differences of feature vectors and classified using gradient boosting. All results reported in this notebook are development results obtained on a 0.1% subset of the PAN 2026 training data. Seven classifiers were compared: LightGBM achieved the strongest validation performance with an F1-score of 0.577, outperforming XGBoost (0.543), Random Forest (0.343), and linear baselines. Linear SVC failed to identify any positive author-switch instances, highlighting the challenge of class imbalance. LightGBM was selected as the planned submission model based on these experiments. The paper focuses on understanding when and why interpretable stylometric features work, rather than on achieving competitive benchmark scores.

Keywords

Author Switch Detection, Style Change Detection, Stylometry, Explainable Machine Learning, Multi-Author Writing Style Analysis

1. Introduction

Authorship analysis is a subfield of natural language processing concerned with identifying, verifying, and profiling authors from the linguistic patterns embedded in their writing. A central task within this field is *style change detection*: given a document that may have been written by more than one person, identify the exact sentence boundaries at which one author's contribution ends and another's begins [1].

The PAN workshop series at CLEF has driven progress on this problem since 2016, evolving from document-level, single versus multi-author classification toward fine-grained sentence-level author switch detection [2]. The 2026 edition [3] focuses on sentence-level detection under three controlled difficulty levels: **easy** (author switches coincide with topic changes), **medium** (some switches occur within the same topic), and **hard** (all text is within a single topic, requiring pure style discrimination).

Recent transformer-based and graph neural network approaches have achieved substantially stronger performance than traditional stylometric methods on PAN-style author-switch detection benchmarks [4, 5]. These models do not, however, explain which stylistic dimensions drive any individual prediction. In forensic linguistics, academic integrity investigations, and similar settings, that kind of explanation is often just as important as the prediction itself.

This paper presents an interpretable system developed for the PAN 2026 shared task at CLEF. The goal is not to outperform neural approaches but to evaluate how well transparent, handcrafted stylometric

CLEF 2026 Working Notes, 9–12 September 2026, Grenoble, France

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features perform under the controlled difficulty conditions of the benchmark, and to analyse clearly where they fail.

Preliminary note. All results reported in this notebook are development results obtained on a 0.1% subset of the PAN 2026 training data. The system has not yet been submitted to TIRA for official evaluation. The reported scores should not be interpreted as benchmark performance. Full-scale experiments and TIRA submission are planned for future work.

The contributions of this work are:

- An interpretable sentence-level author switch detection pipeline using a 12,353-dimensional handcrafted stylometric representation comprising lexical, punctuation, function-word, character n-gram, POS, and language-model features.
- A pairwise absolute-difference feature representation that enables direct attribution of predictions to named stylistic signals.
- A comparative evaluation of seven classifiers under the same feature representation, identifying gradient boosting as the most effective approach on the development subset.
- An analysis of class imbalance effects, demonstrating why accuracy alone is insufficient for evaluating style-change detection systems.

The paper is organised as follows. Section 2 formally defines the task and research questions. Section 3 describes application scenarios. Section 4 describes the dataset. Section 5 reviews related work. Section 6 gives a system overview. Section 7 details the methodology. Section 8 reports development experiments and results. Section 9 discusses the findings. Section 10 concludes.

2. Problem Definition

Let a document D consist of n consecutive sentences:

$$D = \{S_1, S_2, \ldots, S_n\}.$$

For each adjacent pair (S_i, S_{i+1}), the task is to predict a binary label $y_i \in \{0, 1\}$:

$$y_i = \begin{cases} 0 & \text{if } S_i \text{ and } S_{i+1} \text{ were written by the same author,} \\ 1 & \text{if an author switch occurs between } S_i \text{ and } S_{i+1}. \end{cases}$$

The sequence of labels $\{y_1, \ldots, y_{n-1}\}$ fully encodes the authorship structure of the document. From these labels one can recover whether the document is multi-authored and which sentences belong to which author segment. The task is harder at the sentence level than at the paragraph level studied in earlier PAN editions because each segment contains on average only 15–20 words, leaving very little text for stylistic evidence [3].

2.1. Research Questions

1. Which stylometric feature groups—lexical, punctuation, function-word, character n-gram, POS, or language-model—provide the most discriminative signal for sentence-level author switch detection?
2. How do ensemble tree-based classifiers compare with linear models and neural networks when applied to high-dimensional stylometric feature representations?
3. How does severe class imbalance affect classifier behaviour, and which metrics are most informative for model selection?

3. Application Scenarios

The ability to identify where one author's contribution ends and another begins is useful in several practical settings. What connects these applications is the need for decisions that can be explained. A user who is told that a style switch was detected will almost always ask why, and a system that works with named stylometric features can give a direct answer.

Academic integrity. When a submitted assignment shows markedly different writing styles in different sections, the boundary between those styles is evidence that part of the work may not have been written by the student. Systems that can locate that boundary support fair assessment.

Gift authorship prevention. In collaborative research, knowing who wrote which part of a paper matters for credit allocation and funding. Automated boundary detection provides an objective check.

Forensic linguistics. Courts encounter documents whose true authorship is disputed. An expert witness must be able to cite specific linguistic features to support an attribution claim [6]. A black-box system cannot meet this standard.

Anonymous online content. Large platforms host text contributed anonymously. Detecting style switches can help identify coordinated inauthentic behaviour, such as a single actor operating multiple accounts.

4. Dataset

This work uses the PAN 2026 Multi-Author Writing Style Analysis dataset [3], which consists of documents constructed from user-generated online forum posts, structured into three difficulty levels:

- **Easy**: author switches coincide with topic changes; lexical cues reinforce stylistic signals.
- **Medium**: multiple topics appear, but some switches occur within the same topic.
- **Hard**: all sentences in a document address the same topic; switches must be detected from stylistic cues alone.

Each document is pre-segmented into sentences, with a binary label provided for every consecutive sentence pair. The primary evaluation metric is the F1-score over all boundary predictions, which balances precision and recall. Official evaluation is conducted through the TIRA platform [7] using the PAN 2026 train/test split. No test-set labels were used during model development.

The training dataset exhibits severe class imbalance. Approximately 85% of sentence pairs belong to the same-author (negative) class, with only 15% representing genuine author-switch boundaries. This imbalance substantially affects classifier behaviour and is discussed further in Section 8.

5. Related Work

Style change detection sits within the broader field of authorship analysis. This section reviews the main approaches that have been applied to this problem, covering both classical stylometric methods and more recent neural systems.

Bevendorff et al. [8] organise authorship analysis into three principal tasks: author profiling, author attribution, and multi-author writing style analysis. Each relies on stylometric features—quantifiable properties of text that reflect an author's writing habits, such as sentence length, punctuation usage, vocabulary richness, and function-word frequencies.

Author attribution is the problem of determining which candidate wrote a disputed text. Authorship attribution has long been studied using stylometric analysis, with classic examples including investigations of the Federalist Papers and other disputed texts. More recently, Jockers and Witten [9] provided a comparative study of machine learning methods for this task. Ramnial et al. [10] showed that attribution can be reframed as pairwise binary verification—if two adjacent segments differ stylistically, a switch is inferred—which motivates the pairwise representation used here.

Classical machine learning. Singh et al. [11] extracted character n-grams, POS-tag n-grams, function words, and vocabulary richness, then applied Logistic Regression on pairwise absolute feature differences, achieving F1 0.657 on PAN 2021 paragraph-level data, providing a useful reference point for handcrafted stylometric approaches. The present work extends this approach to the sentence level with a substantially larger feature set. Rios-Toledo et al. [12] found that higher-order n-grams ($n = 3$–5) and POS-tag sequences outperform unigrams for detecting style changes in literary text, and that POS features are especially robust to topic shifts. Alshamasi and Menai [13] applied ensemble K-means clustering to stylistic feature vectors, reaching F1 0.49 at the sentence level, establishing a useful unsupervised reference point.

Transformer-based and neural methods. Neural approaches currently achieve the highest reported performance on PAN benchmarks. Chen et al. [4] proposed a writing style embedding based on contrastive learning for multi-author writing style analysis. Alsheddi and Menai [5] applied graph convolutional networks to the task. These systems are not directly compared here because they differ substantially in architecture, training scale, and computational requirements. They are noted as context for the difficulty of the task.

Gap. Existing transformer-based systems optimise for performance but offer no per-prediction account of which stylistic dimensions drove the result. To the best of our knowledge, limited prior work has examined handcrafted feature importance under the sentence-level difficulty conditions introduced in PAN 2026. This gap motivates the present work.

6. System Overview

The proposed system follows a modular stylometric pipeline consisting of feature extraction, pairwise representation learning, and supervised classification. The design emphasises interpretability while supporting a wide range of feature types and machine-learning models.

Stage 1 – Preprocessing. Each document arrives pre-segmented, with one sentence per line. Sentence boundaries are therefore identified by splitting on newline characters. Words within each sentence are split on whitespace for feature computation. Punctuation, stop words, and special characters are retained rather than removed, since these elements carry author-specific signal.

Stage 2 – Feature extraction. A 12,353-dimensional stylometric representation is extracted from every sentence, comprising lexical, punctuation, function-word, character n-gram, part-of-speech, and language-model features. Each sentence is represented by the concatenation of all six feature groups. Character n-gram features constitute the largest group and are generated using a hashing-based representation to maintain a fixed-dimensional feature space.

Stage 3 – Pairwise representation. Author-switch detection is formulated as a boundary-classification problem between adjacent sentences. For each consecutive sentence pair (S_i, S_{i+1}), a pairwise feature vector is computed as the element-wise absolute difference of the two sentence feature vectors:

$$\mathbf{x}_i = \left|\mathbf{f}(S_i) - \mathbf{f}(S_{i+1})\right| \in \mathbb{R}^k.$$

A large value in dimension j means the two sentences differ strongly on feature j. The classifier then learns which feature differences are most predictive of a switch.

Stage 4 – Classification. The pairwise vectors are passed to supervised classifiers. The framework supports Logistic Regression, Linear SVC, Multi-Layer Perceptron, Random Forest, Extra Trees, XGBoost, and LightGBM. Based on development experiments, LightGBM was selected as the planned submission model.

The full pipeline runs on CPU only. TIRA submission is planned for future work.

7. Proposed Methodology

This section describes each pipeline stage in more detail. The design choices were guided by three priorities: interpretability (each feature has a clear linguistic motivation), computational feasibility (the system must run on a low-resource CPU machine), and compatibility with the pairwise representation used by Singh et al. [11].

7.1. Preprocessing

Each document in the PAN 2026 dataset is pre-segmented, with one sentence per line. Sentence boundaries are therefore extracted by splitting on newline characters directly, without the need for a sentence tokenization library. Words within a sentence are split on whitespace. Following the observation of Zamir et al. [14], stop words, punctuation, and special characters are retained during feature extraction rather than removed, since these elements carry author-specific signal.

7.2. Feature Extraction

A comprehensive stylometric feature set is extracted from every sentence. Features are organised into six groups to support interpretability and feature-group analysis. Table 1 summarises the feature groups and their dimensionalities.

Table 1
Overview of the six stylometric feature groups extracted per sentence, with exact feature counts per group.

Feature Group	Description	Features
Lexical	Sentence length, vocabulary richness, word-length statistics	11
Punctuation	Counts and ratios of punctuation and special symbols	14
Function Words	Function-word and pronoun usage statistics	4
Character N-grams	Hashed character bi-, tri-, and four-grams	12,288
POS Features	POS tag distributions and POS bigrams	27
N-gram Language Model	Entropy, perplexity, and average log-probability for $n = 1, 2, 3$	9
Total		**12,353**

In total, each sentence is represented using 12,353 stylometric features. Character n-gram features constitute the largest feature group and are generated using a hashing-based representation to maintain a fixed-dimensional feature space.

Group 1: Lexical features. Sentence length (character count, word count, token count), average word length, type-token ratio, hapax legomena ratio, and word-length distributions.

Group 2: Punctuation features. Per-sentence counts (normalised by sentence length) of commas, periods, semicolons, quotation marks, question marks, exclamation marks, parentheses, dashes, and related symbols.

Group 3: Function-word features. Normalised frequencies of a function-word list (prepositions, articles, conjunctions, auxiliary verbs) and the overall stop-word ratio. Function words are largely unconscious choices and are theoretically more robust to topic shifts than content words [6].

Group 4: Character n-gram features. Character 2-, 3-, and 4-gram frequencies are extracted using a hashing-based representation to capture spelling preferences, formatting habits, and subword stylistic patterns.

Group 5: Part-of-speech features. POS-tag distributions and POS bigrams provide a lightweight representation of syntactic preferences and grammatical structure.

Group 6: N-gram language model features. Entropy, perplexity, and average log-probability for $n = 1, 2, 3$ are computed from sentence-level language models. These statistics have been used in prior work as indicators of sentence-level linguistic predictability and fluency patterns that may distinguish individual authors [14].

The final sentence representation is obtained by concatenating all six feature groups into a single 12,353-dimensional stylometric vector.

7.3. Pairwise Representation

The key modelling assumption is that an author switch between S_i and S_{i+1} will produce a measurable difference in their feature vectors. Let $\mathbf{f}(S_i) \in \mathbb{R}^k$ be the feature vector for sentence S_i. The pairwise representation is:

$$\mathbf{x}_i = \left|\mathbf{f}(S_i) - \mathbf{f}(S_{i+1})\right| \in \mathbb{R}^k,$$

where the absolute value is taken element-wise. A large value in dimension j means the two sentences differ strongly on feature j. This design, used by Singh et al. [11] at the paragraph level, directly supports post-hoc explanation of any individual prediction.

7.4. Classifiers

Seven supervised learning algorithms were evaluated under the same stylometric representation.

Logistic Regression (LR). A linear classifier whose learned coefficients indicate the contribution of each pairwise feature difference.

Linear Support Vector Machine (SVC). A margin-based linear classifier. A linear kernel is used for computational feasibility on the large feature space.

Multi-Layer Perceptron (MLP). A feed-forward neural network implemented using scikit-learn's `MLPClassifier` with two hidden layers containing 128 and 64 neurons respectively, ReLU activation, Adam optimisation, adaptive learning-rate scheduling, and early stopping.

Random Forest (RF). An ensemble of decision trees trained using balanced class weighting.

Extra Trees (ET). An extremely randomised tree ensemble.

XGBoost (XGB). Gradient-boosted decision trees.

LightGBM (LGBM). Histogram-based gradient boosting, selected as the planned submission model based on development experiments.

7.5. Experimental Setup

During system development, a 0.1% subset of the PAN 2026 training corpus was used to perform model comparison and hyperparameter exploration. Five-fold cross-validation was applied to this subset to evaluate candidate classifiers. Model selection was based on the mean validation F1-score across folds. All random seeds were fixed at 42.

The system has not yet been submitted to TIRA for official evaluation. The results reported in Section 8 are development results only and should not be interpreted as benchmark performance. LightGBM was selected as the planned submission model based on these experiments. TIRA submission and full-scale training are planned for future work.

Table 2
Key reproducibility parameters.

Parameter	Value
Total Features	12,353
Pairwise Mode	Absolute Difference
Development Subset	0.1% of training data
Cross-Validation	5-fold
Random Seed	42
Selected Model	LightGBM
TIRA Submission	Not yet submitted

8. Development Experiments and Results

All results reported in this section are development results obtained on a 0.1% subset of the PAN 2026 Multi-Author Writing Style Analysis training dataset. These experiments were intended for model comparison and selection rather than final system evaluation.

8.1. Class Distribution

The training dataset exhibits severe class imbalance, with approximately 85% of sentence pairs belonging to the same-author (negative) class and only 15% representing genuine author-switch boundaries. This imbalance has significant implications for classifier behaviour: models can achieve high accuracy by simply predicting the majority class, making accuracy an unreliable metric for model selection. The F1-score, which balances precision and recall, provides a more informative evaluation criterion for this task.

8.2. Classifier Comparison

Table 3 reports five-fold cross-validation results for all evaluated classifiers on the development subset. Performance is reported using accuracy, precision, recall, and F1-score for the author-switch (positive) class.

Table 3
Five-fold cross-validation results on a 0.1% subset of the PAN 2026 training data. Metrics reported for the author-switch (positive) class. These are development results only.

Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.760	0.312	0.252	0.279
Linear SVC	0.815	0.000	0.000	0.000
MLP	0.787	0.279	0.103	0.148
Random Forest	0.837	0.771	0.253	0.343
Extra Trees	0.824	0.633	0.104	0.165
XGBoost	0.874	0.832	0.405	0.543
LightGBM	**0.874**	0.773	**0.463**	**0.577**

LightGBM achieved the strongest overall performance with a mean F1-score of 0.577 and accuracy of 87.4%, slightly outperforming XGBoost, which achieved an F1-score of 0.543. Ensemble tree-based approaches consistently outperformed linear and neural models, suggesting that they are better suited for capturing complex interactions among the high-dimensional stylometric feature groups used in this work.

8.3. Impact of Class Imbalance

Although Linear SVC achieved relatively high accuracy (81.5%), it failed to identify any positive author-switch instances, resulting in zero precision, recall, and F1-score. This behaviour reflects the severe class imbalance present in the dataset: the model learned to predict only the majority class. This demonstrates why accuracy alone is insufficient for evaluating style-change detection systems and highlights the importance of using metrics that account for both classes, particularly when the positive class is of primary interest.

The performance gap between linear models and ensemble methods also illustrates the difficulty of learning decision boundaries in the 12,353-dimensional pairwise feature space under class imbalance. Gradient boosting approaches (LightGBM and XGBoost) were most effective at identifying author-switch boundaries, achieving recall values of 0.463 and 0.405 respectively, substantially higher than all other methods.

8.4. Model Selection

Based on the development experiments, LightGBM was selected as the planned submission model, achieving the highest validation F1-score of 0.577. TIRA submission and training on the complete dataset are planned for future work.

A detailed feature importance analysis across the six stylometric feature groups was not conducted as part of the current development experiments. This analysis is planned for future work and will provide a transparent account of which stylistic feature groups contribute most to author-switch detection performance under the controlled difficulty conditions of the PAN 2026 benchmark.

9. Discussion

This section discusses the methodological findings and limitations of the current approach based on the development experiments.

9.1. Effectiveness of Ensemble Methods

The development results clearly demonstrate that ensemble tree-based methods, particularly gradient boosting approaches, are substantially more effective than linear models for this task. LightGBM and XGBoost achieved F1-scores of 0.577 and 0.543 respectively, compared to 0.279 for Logistic Regression. This performance gap suggests that the decision boundary between same-author and author-switch sentence pairs involves complex, nonlinear interactions among stylometric features that linear models cannot capture.

9.2. The Challenge of Class Imbalance

The failure of Linear SVC—achieving zero recall despite 81.5% accuracy—provides a concrete illustration of why accuracy is an inadequate metric for imbalanced style-change detection. With only 15% of sentence pairs representing genuine author switches, a model that predicts "no switch" for every boundary would achieve 85% accuracy while being completely useless for the task. The F1-score, which penalises both false positives and false negatives, provides a more meaningful evaluation criterion.

9.3. Limitations

Several limitations of the current work should be noted. First, all reported results were obtained on only 0.1% of the training data; performance on the full dataset may differ substantially. Second, the system has not yet been submitted to TIRA for official evaluation, so no test-set results are available. Third, the pairwise representation treats each sentence pair independently, discarding document-level context that could help distinguish genuine author switches from quotations or temporary style shifts. Fourth, class imbalance was addressed only through balanced class weighting in the tree-based models; more sophisticated approaches such as oversampling or cost-sensitive training may yield further improvements.

9.4. Interpretability Considerations

A key advantage of the handcrafted feature approach is that every prediction can be traced back to named stylistic dimensions. Unlike transformer-based systems that operate on opaque embeddings, the pairwise absolute-difference representation enables direct inspection of which feature groups contribute most to any individual boundary prediction. Although a formal feature importance analysis was not included in the current experiments, the framework is designed to support such analysis in future work.

10. Conclusion and Future Work

This paper presented an interpretable stylometric system for sentence-level author switch detection, developed for the PAN 2026 shared task at CLEF. The system uses a 12,353-dimensional handcrafted feature representation spanning lexical, punctuation, function-word, character n-gram, POS, and language-model features. Consecutive sentence pairs are represented as absolute differences and classified using gradient boosting.

All results reported in this notebook are development results obtained on a 0.1% subset of the PAN 2026 training data. Seven classifiers were compared under five-fold cross-validation. LightGBM achieved the strongest performance with an F1-score of 0.577, substantially outperforming linear baselines. The experiments also highlighted the critical impact of class imbalance: Linear SVC achieved 81.5% accuracy while failing to identify any author-switch boundaries, demonstrating why accuracy alone is insufficient for this task.

The main contribution of this work is a systematic evaluation of classical stylometric features combined with modern gradient boosting for sentence-level author switch detection, providing a transparent and reproducible development baseline.

Future work includes: submitting the system to TIRA for official evaluation on the PAN 2026 test set; conducting full-scale training on the complete dataset; performing detailed feature importance analysis across the easy, medium, and hard difficulty levels; incorporating document-level context to improve switch detection; and applying more sophisticated class imbalance mitigation strategies.

Acknowledgments

The authors thank the PAN 2026 organisers for providing the benchmark dataset and evaluation framework. The authors also acknowledge the University of Johannesburg for institutional support.

Declaration on Generative AI

During the preparation of this work, the authors used ChatGPT for grammar and spelling assistance and for improving the clarity of written text. All generated content was reviewed and edited by the authors as needed. The authors take full responsibility for the content of this publication.

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